

# Branch Specialization in Attention

## Toy Local-vs-Induction Competition Checkpoint

Research checkpoint

May 22, 2026

**Stronger RQ5 test:** Does heterogeneous head dimension produce a stable role partition under task competition?

Desired positive result:

small heads  $\rightarrow$  local role,    large head  $\rightarrow$  global induction role

This clean result did not appear.

# Task

Each sequence has two scored regions.

Local:

$$[x, \text{SEP}, x]$$

At SEP, predict the immediately previous token  $x$ .

Induction:

$$[y_1, \dots, y_{16}, y_1, \dots, y_{16}]$$

At the second  $y_i$ , predict  $y_{i+1}$ .

# Experimental Arms

| Config   | Head dimensions | Purpose                    |
|----------|-----------------|----------------------------|
| uniform4 | [32,32,32,32]   | same-head-count baseline   |
| hetero4  | [16,16,32,64]   | heterogeneous intervention |
| uniform2 | [64,64]         | fewer/wider control        |
| 64-first | [64,16,16,32]   | position control           |

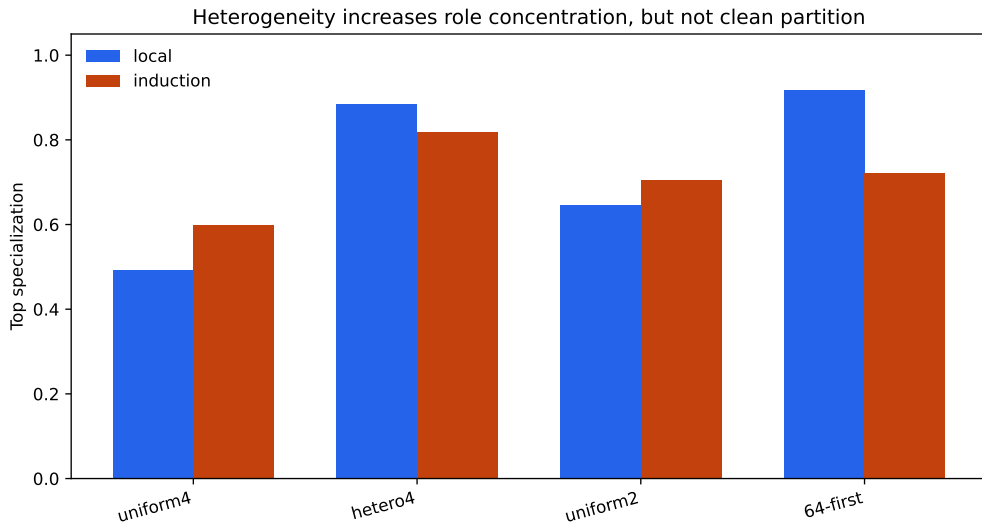
All models use two layers,  $d_{model} = 128$ , five seeds, and total attention dimension 128.

# Both Objectives Learn

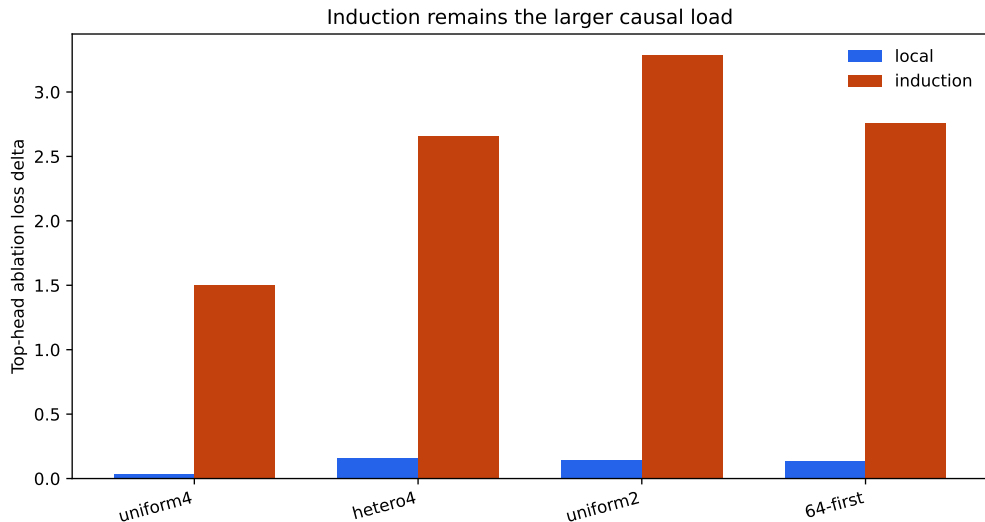
| Config   | Local acc. | Induction acc. |
|----------|------------|----------------|
| uniform4 | 1.0000     | 0.9999         |
| hetero4  | 1.0000     | 0.9992         |
| uniform2 | 1.0000     | 0.9976         |
| 64-first | 0.9999     | 0.9986         |

The role comparison is not primarily a task-failure artifact.

# Role Concentration

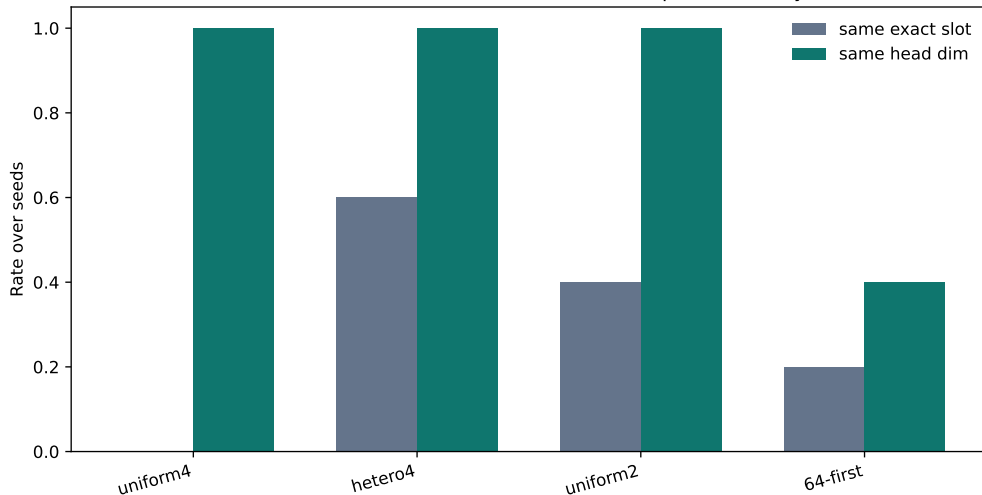


# Causal Load



# Role Overlap

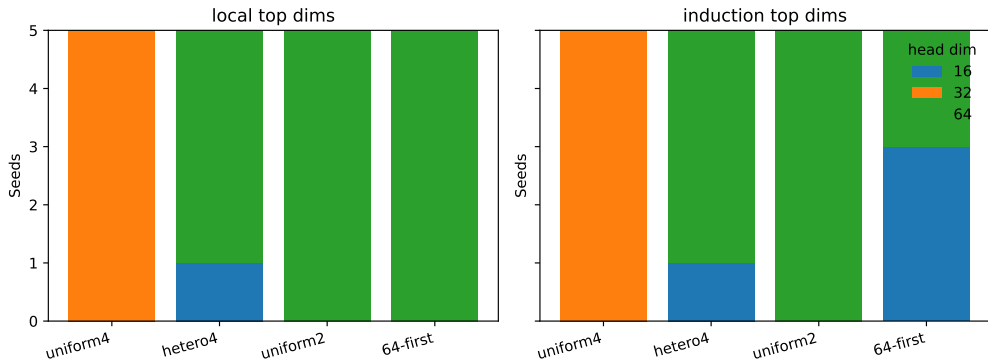
Local and induction roles often do not separate cleanly





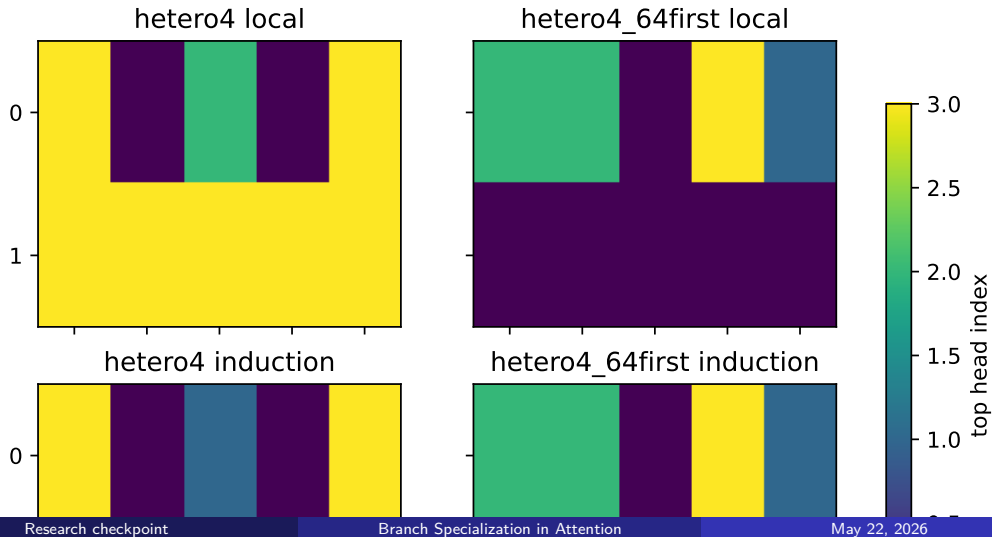
# Top Dimensions

Top role dimensions across seeds



# Layout Sensitivity

Competition makes role assignment layout-sensitive



Still supported:

- heterogeneous head dimensions increase causal specialization concentration;
- heterogeneous slots break permutation symmetry relative to uniform4.

Not supported yet:

- small heads reliably become local heads;
- the large head reliably becomes the global induction head.

Narrow the Stage 3 claim:

**Heterogeneity is a permutation-symmetry-breaking stabilizer, not a guaranteed semantic role allocator.**

Next:

- run more heterogeneous permutations;
- report structure-aware slot stability;
- treat layout sensitivity as a real constraint on the intervention story.

- Script: `scripts/toy_competition_head_dim_intervention.py`
- Output: `results/phase3_toy_competition_head_dim_intervention`
- Report: `doc/phase3_toy_competition_head_dim_intervention.md`
- Data copied into this deck folder under `data/`